

Name: _____



Recognising geometric sequences

Task A: Finding the common ratio

1) Divide each term by the previous term to find the common ratio in each of these geometric sequences.

a) 15, 105, 735, 5145, 36 015, ...

b) 0.08, 0.72, 6.48, 58.32, 524.88, ...

c) 567, 189, 63, 21, 7, ...

d) $\frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \frac{27}{32}, \frac{81}{64}, \dots$

2) a) This sequence is geometric. Show that it is and write a sentence to justify why it is.

768, 576, 432, 324, 243, ...

b) This sequence is not geometric. Show that it is not and write a sentence to justify why it is not.

756, 252, 126, 42, 21, ...

Task B: Using the common ratio

1) Find the missing terms in these geometric sequences.

a) $\boxed{}, \boxed{}, 51, 153, \boxed{} \dots$ b) $\boxed{}, 225, 45, \boxed{}, \boxed{} \dots$

c) $\boxed{}, 0.14, 2.8, \boxed{}, \boxed{} \dots$ d) $\boxed{}, \boxed{}, \frac{28}{27}, \frac{56}{81}, \boxed{} \dots$

2) Find the two possible geometric sequences given these missing, non-successive terms.

1 st	2 nd	3 rd	4 th	5 th
4	$\boxed{}$	100	$\boxed{}$	$\boxed{}$